

```
In [ ]: 1.String:Immutable,Ordered,Duplicates items are allowed,' ',' ',' ',' ',' ',' ',' '
2.List:Mutable,Ordered,Duplicates items are allowed,[]
3.Tuple:Immutable,Ordered,Duplicates items are allowed,()
4.Set:Mutable,Unordered,Duplicates items are not allowed,set(),{}
```

Set data Types:

```
In [ ]: Properties:
1)Mutable>> Changable(Add, delete ,update)
2)It is in Unordered(We can not use indexing and slicing )
3)Duplicate Items are not allowed.
4)It is enclosed by curly brackets {}
5)initialize with set()
6)Set items are comma separated.
```

```
In [2]: set1={10,20,30,40,50}
set1
```

```
Out[2]: {10, 20, 30, 40, 50}
```

```
In [3]: set1[2]
```

```
-----
TypeError                                Traceback (most recent call last)
Cell In[3], line 1
----> 1 set1[2]

TypeError: 'set' object is not subscriptable
```

```
In [4]: list1=[10,20,30,40,50]
list1[2]
```

```
Out[4]: 30
```

```
In [5]: set1={10,20,30,40,'python','machine','learning'}
set1
```

```
Out[5]: {10, 20, 30, 40, 'learning', 'machine', 'python'}
```

```
In [6]: set1={10,20,30,40,'python',20,'machine','learning'}
set1
```

```
Out[6]: {10, 20, 30, 40, 'learning', 'machine', 'python'}
```

```
In [8]: set1={10,20,30,40,'python',20,'PYTHON','machine','learning'}
set1
```

```
Out[8]: {10, 20, 30, 40, 'PYTHON', 'learning', 'machine', 'python'}
```

```
In [9]: a=[]  
        type(a)
```

Out[9]: list

```
In [10]: a=list()  
         type(a)
```

Out[10]: list

```
In [11]: b=()  
         type(b)
```

Out[11]: tuple

```
In [12]: c=""  
         type(c)
```

Out[12]: str

```
In [13]: c=str()  
         type(c)
```

Out[13]: str

```
In [14]: y=set()  
         type(y)
```

Out[14]: set

```
In [16]: s1={3,4,5,6,7.5,'python'}  
         s1
```

Out[16]: {3, 4, 5, 6, 7.5, 'python'}

```
In [17]: s1={3,4,5,6,7.5,'python',(10,20,30)}  
         s1
```

Out[17]: {(10, 20, 30), 3, 4, 5, 6, 7.5, 'python'}

```
In [18]: s1={3,4,5,6,7.5,'python',(10,20,30),[1,2,3]}  
         s1
```

```
-----  
TypeError                                Traceback (most recent call last)  
Cell In[18], line 1  
----> 1 s1={3,4,5,6,7.5,'python',(10,20,30),[1,2,3]}  
      2 s1
```

TypeError: unhashable type: 'list'

Type casting

```
In [ ]: list>>tuple
list>>set

tuple>>list
tuple>>set

set>>list
set>>tuple
```

```
In [19]: x=[2,3,4,5]
y=tuple(x)
y
```

```
Out[19]: (2, 3, 4, 5)
```

```
In [20]: x=[2,3,4,5]
y=set(x)
y
```

```
Out[20]: {2, 3, 4, 5}
```

```
In [21]: s1={2,3,4,5,6,7}
x=list(s1)
y=tuple(s1)
print(x)
print(y)
```

```
[2, 3, 4, 5, 6, 7]
(2, 3, 4, 5, 6, 7)
```

Access Set Items

```
In [22]: s1={2,3,4,5,6,7}
for i in s1:
    print(f"i=={i}")
```

```
i==2
i==3
i==4
i==5
i==6
i==7
```

```
In [24]: s1={10,20,'Machine Learning',"Pune",5.67,100}
print(s1)
for i in s1:
    print(i)
```

{'Machine Learning', 100, 5.67, 20, 10, 'Pune'}

Machine Learning

100

5.67

20

10

Pune

```
In [25]: s1={10,20,'Machine Learning',"Pune",5.67,100}
for i,val in enumerate(s1):
    print(val)
```

Machine Learning

100

5.67

20

10

Pune

Set Functions

```
In [ ]: 1)add()
2)update()
3)union()

4)remove()
5)discard()
6)pop()
7)clear()

8)intersection()
9)intersection_update()

10)difference()
11)difference_update()

12)symmentric_difference
13)symmetric_difference_update()

14)issuperset()
15)issubset()
16)isdisjoint()
```

1.add()

```
In [ ]: It is used to add only one item in set.
Set.add(Value)
```

```
In [26]: s1={3,4,1,6,7,78,10,2,3,4}
s1.add(100)
s1
```

Out[26]: {1, 2, 3, 4, 6, 7, 10, 78, 100}

```
In [27]: s1={2,4,5,6,7,8,2,3}
s1.add('python')
s1
```

Out[27]: {2, 3, 4, 5, 6, 7, 8, 'python'}

```
In [28]: s1={2,4,5,6,7,8,2,3}
s1.add(6)
s1
```

Out[28]: {2, 3, 4, 5, 6, 7, 8}

```
In [29]: s1={"pune","mumbai","delhi"}
s1.add("chennai")
s1
```

Out[29]: {'chennai', 'delhi', 'mumbai', 'pune'}

```
In [30]: s1='python'
x=s1.replace('python','java')
x
```

Out[30]: 'java'

2)update()

```
In [ ]: Set.update(iterable)
Iterable>>Set,list,tuple
```

```
In [31]: x=[2,3,4,5]
y=[3,10,20,30]
x.extend(y)
x
```

Out[31]: [2, 3, 4, 5, 3, 10, 20, 30]

```
In [32]: x={2,3,50,60,6,7}
y={10,20,30,40,50,30,40}
y.update(x)
x
```

Out[32]: {2, 3, 6, 7, 50, 60}

```
In [33]: x={2,3,50,60,6,7}
y={10,20,30,40,50,30,40}
y.update(x)
y
```

Out[33]: {2, 3, 6, 7, 10, 20, 30, 40, 50, 60}

```
In [34]: x={2,3,50,60,6,7}
y={10,20,'30',40,50,30,40}
x.update(y)
x
```

Out[34]: {10, 2, 20, 3, '30', 30, 40, 50, 6, 60, 7}

```
In [35]: x={2,3,4,5,6}
y=[10,20,30,40,50]
x.update(y)
x
```

Out[35]: {2, 3, 4, 5, 6, 10, 20, 30, 40, 50}

```
In [36]: x={2,3,4,5,6}
y=[10,20,30,40,50]
y.update(x)
y
```

AttributeError

Traceback (most recent call last)

Cell In[36], line 3

```
1 x={2,3,4,5,6}
2 y=[10,20,30,40,50]
----> 3 y.update(x)
4 y
```

AttributeError: 'list' object has no attribute 'update'

```
In [37]: x={2,3,4,5,6}
y=[10,20,30,40,50]
z=('python','python','data science')
x.update(z)
x
```

Out[37]: {2, 3, 4, 5, 6, 'data science', 'python'}

```
In [38]: x={2,3,4,5,6}
y=[10,20,30,40,50]
z=('python','python','data science')
z.update(x)
z
```

```
-----
AttributeError                                Traceback (most recent call last)
Cell In[38], line 4
      2 y=[10,20,30,40,50]
      3 z=('python','python','data science')
----> 4 z.update(x)
      5 z

AttributeError: 'tuple' object has no attribute 'update'
```

3)union()

```
In [ ]: Return a set containing the union of sets
New_set=Set.union(iterable)
ietrable>>set,tuple,list
```

```
In [43]: s1={3,4,5,6,7,8,5,3}
s2={10,20,30,40,50,20,30}
s1.update(s2)
print("set 1:",s1)
print("Set 2:",s2)
```

```
set 1: {3, 4, 5, 6, 7, 8, 40, 10, 50, 20, 30}
Set 2: {50, 20, 40, 10, 30}
```

```
In [41]: s1={3,4,5,6,7,8,5,3}
s2={10,20,30,40,50,20,30}
new_set=s1.union(s2)
print("Set 1:",s1)
print("Set 2:",s2)
print("New Set:",new_set)
```

```
Set 1: {3, 4, 5, 6, 7, 8}
Set 2: {50, 20, 40, 10, 30}
New Set: {3, 4, 5, 6, 7, 8, 40, 10, 50, 20, 30}
```

```
In [ ]:
```